MIL-H-43925D
27 April 1988
SUPERSEDING
MIL-H-43925C
30 September 1986
(See 6.7)

MILITARY SPECIFICATION

HELMET, FLYER'S, PROTECTIVE, SPH-4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the SPH-4 flyer's protective helmet.

* 1.2 Classification. The helmets shall be of the following sizes (see 6.2).

Sizes: Small
Regular
X-Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
DDD-L-20 - Label: For Clothing, Equipage, and Tentage (General Use)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 10760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-T-60 - Tape, Packaging, Waterproof
PPP-T-76 - Tape, Pressure Sensitive Adhesive, Packaging/Paper (For Carton Sealing)

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-L-35078 - Loads, Unit· Preparation of Semiperishable Subsistence Items, Clothing, Personal Equipment and Equipage; General Specification For
MIL-V-43511 - Visor, Flying Helmet, Polycarbonate
MIL-A-46106 - Adhesive-Sealants, Silicone, RTV, General Purpose
MIL-H-49225 - Headset Microphone Kit MK 896A/AIC (Components of)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2.1.2 Other Government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

JOINT COMMITTEE ON PRINTING (JCP) PUBLICATIONS

Government Paper Specification Standards

(Copies may be obtained from the Superintendent of Documents, U S. Government Printing Office, Washington, DC 20402)

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

8-2-481 - Helmet, Flyer's, Protective, SPH-4; Spacer, Visor Housing
8-2-523 - Visor Lock Assembly
8-2-524 - Packaging of Helmets
8-2-800 - Helmet, Flyer's, Protective, SPH-4 Assemblies

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8-2-803 - Shell, Small/Regular
8-2-804 - Shell, X-Large
8-2-812 - Earcup Retainer Assembly
8-2-814 - Earcup Retention Cross Strap Assembly
8-2-815 - Chinstrap Assembly
8-2-817 - Post Slotted
8-2-818 - Post
8-2-821 - Clip
8-2-825 - Testing of Chinstrap (When Procured Separately)
8-2-827 - Pad
8-2-832 - Testing of Earcup Retainer Eyelet Chape
8-2-833 - Testing of Earcup Retainer (When Procured Separately)
8-2-834 - Hook
8-2-835 - Clip
8-2-839 - Body Pattern
8-2-840 - Ring Assembly Reinforcement
8-2-841 - Extension, Body
8-2-853 - Pad Assembly
8-2-855 - Body
8-2-856 - Slide
8-2-857 - Cushion Backing
8-2-858 - Retainer
8-2-859 - Spacer
8-2-860 - Cushion, Inner
8-2-862 - Cushion, Earpad
8-2-863 - Button, Earpad
8-2-864 - Ring, Connector
8-2-866 - Cover Earpad
8-2-897 - Suspension System, Form Fit

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

S12.6/ASA 55 - Method For Measurement of the Real-Ear Attenuation of Hearing Protectors
Z90.1b-1979 - Specification for Protective Headgear for Vehicular Users

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 First article When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4)

3.2 Samples

3.2.1 Guide samples. Samples of the helmet or components, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from the document may appear in the samples, in which case the document shall govern

3.2.2 Standard color samples Unless otherwise specified, standard samples for color shall apply (see 6.3)

3.3 Materials and components. Materials and components shall be as specified on the applicable drawings and as specified herein (see 6.6). For materials or components for which it is stated "or equal", if the contractor proposes to use an item considered to be equal to the material or component specified, prior to its use the contractor shall furnish a sample of the material or component, with supporting data to the contracting officer for subsequent evaluation by the responsible military agency The supporting data required shall prove the functional equivalence and design compatibility of the item proposed to be used.

3.3.1 Adhesives The adhesives shall be applied to the helmet components in a manner that will not degrade the liners or affect the finish of any helmet component.

3.3.1.1 Rubber edging The adhesive for bonding the rubber edging of the helmet shall meet the bonding requirements specified in 3.5.3 and 3.5.4.

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3.3.2 Instruction manual. An instruction manual shall be furnished by the contractor with each helmet The manual shall be placed into the suspension of the completed helmet. The printing, collation, folding, and stitching of the manual shall be in accordance with the standard sample (see 6.3). The cover shall be antique white book paper conforming to JCP-A-100, 140 pound weight basis 25 by 38 inches per 1000 sheets (see 4.4.1.1). The text paper shall be white offset book paper conforming to JCP-A-60, 120 pound weight basis 25 by 38 inches per 1000 sheets (see 4.4.1 1)

3.3.3 Identification label Each helmet shall have a permanent type identification label applied in a permanent manner to the interior surface of the helmet shell located to the rear of the right ear cavity adjacent to the cut out in the helmet liner. The label shall be white, approximately 1 by 2 inches in size and the characters shall be black, 3/32 inch high minimum The label shall have the following information.

Nomenclature
Size
Stock No.
Contract No.
Lot No.
Contractor Name

3.3.3 1 Identification labels and markings on components., When specified on the applicable drawings, components shall have identification labels or markings, as applicable Unless otherwise specified on the applicable drawing, the identification label for textile components shall conform to type IV, class 5 of DDD-L-20, except the lot number shall be included in the label contents. For components other than textile, the identification label or marking shall be as specified on the applicable drawing.

* 3.4 Design and construction. The design and construction of the helmet shall be as specified herein, and as shown on Drawing 8-2-800 and all subsidiary drawings and parts lists pertaining thereto.

3.4.1 Stitching. All stitching shall be as specified on the applicable drawings and as specified herein.

3.4.2 Thread tension. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.3.3 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When a thread break or bobbin run-out occurs during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of tne stitching. 1/

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b. Any thread break or two or more consecutive skipped or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.4 Automatic stitching Automatic machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met, and at least three or more tying, overlapping or back stitches are used to secure the ends of the stitching

3 4 5 Thread ends. All thread ends shall be trimmed to a length of not more than 1/4 inch

3 4 6 Setting of eyelets and grommets A hole shall be prepunched to receive the eyelet and grommet The hole shall be smaller than the outside diameter of the eyelet or grommet barrel so that the barrel must be forced through the hole. The eyelets and grommets installed in the chinstrap and earcup retainer shall meet the performance requirements specified in 3 5 9, 3 5.10, and 3.5.11. The hole shall not be punched in the setting operation with the eyelet or grommet barrel The eyelets and grommets shall be securely clinched without splitting and in a manner that will prevent rotation of the eyelets and grommets freely within the webbing, detachment of the eyelets and grommets from the webbing, or cutting of the adjacent material (see 3 5 9 and 3.5.10). The minimum inside diameter of the eyelet hole after clinching shall be as specified on the applicable drawing Unless otherwise specified, the washer may be on either side of the components on which installed.

3.4.7 Setting of snap fasteners A hole shall be prepunched to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials, such that the ends of the barrels of the button and eyelet components are completely rolled back (see 3 5 9 and 3.5.10).

3.4.8 Fusing of ends of nylon webbing. Unless otherwise specified, all ends of nylon webbing shall be fused. The apparatus used to fuse the webbing ends shall be capable of providing sufficient heat to provide a smooth edge with the cut ends of the webbing yarns all fused together. Fusing of the webbing ends shall be accomplished prior to being assembled for stitching.

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3.4.9 Dipping of ends of elastic webbing. All ends of elastic webbing shall be dipped in a melted mixture of 50 percent beeswax and 50 percent paraffin to a depth of 3/16 ± 1/16 inch Temperature of the melted mixture shall be 180° ± 20°F. The dipped ends shall have a smooth and even appearance.

3.4.10 Finish

3.4.10 1 Hardware. Hardware components specified to be finished black on the applicable drawings shall be uniformly coated and free from flaking.

3 4.10.2 Helmet shell. The entire exterior surface of the helmet shell shall be coated with the coating specified on Drawing 8-2-800. The coat-ing shall be either air-dried or force-dried prior to attachment of rubber edging and screws. The dried finish shall be uniform in color and luster over the entire exterior surface of the shell. The final finish shall be free of sags, runs, wrinkles, blistering, or other defects either of improper application or of cure of the coating

3.4.11 Assembly of helmet. The helmet and helmet components shall be assembled as specified on the applicable drawings and as specified below. The contractor is not required to follow the assembly instructions in the exact sequence shown.

3.4.11.1 Visor assembly. The front edge of the visor housing shall not project below the lower bound edge of the helmet. The space between the visor and the housing shall be sufficient to enable the visor to be raised and lowered without touching the housing except at the left and right edges 3/4 inch maximum from edges of visor. If interface does occur, the specified spacers shall be used.

3.4.11.2 Communications assembly

a. Use headset microphone kit, MK 896A/AIC of MIL-H-49225
b. Insert female connector into connector retainer clip with bottom edge connector even with bottom edge of retainer clip. Clinch prongs on retainer clip securely around connector.
c. Attach connector retainer clip on helmet shell using retainer clip plate and mounting screws as shown on Drawing 8-2-800.
d. Attach the cord-strain relief clip to helmet shell as shown in view E, Drawing 8-2-800.
e. Route microphone wire through helmet shell from inside to outside as shown on Drawing 8-2-800. Insert grommet inclosing wire into helmet shell and seat fully.
f. Position shrinkable tubing over connector and retainer clip and apply sufficient heat to shrink tubing firmly around connector and retainer clip.
g. Install earcups in earcup retainer.

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h When installing suspension band assembly, route wires between suspension assembly and liner just above the suspension screws, as shown in the rear view and top view of Drawing 8-2-800

i Attach microphone boom on helmet shell as shown on Drawing 8-2-800. Insert male connector into female connector.

3.5 Performance.

3.5.1 Impact resistance When tested as specified in 4 4 7, the helmet shall not exceed the failure criteria (300 g's) specified in paragraph 9 2.1 of American National Standard Z90.1b-1979 when subjected to impacts with terminal velocities of 19.6 + 30 - 00 feet per second. Helmets shall be impacted after exposure to ambient temperature, low temperature, high temperature, and water immersion

3.5 2 Adhesion of the shell edge binding before aging The edging shall remain firmly bonded to the shell when tested in accordance with 4 4 7. Unbonded areas up to a total of 2 inches in length are acceptable providing no individual unbonded area is more than 1/2 inch in length and 1/8 inch in width There shall be a minimum distance of 1/2 inch between any two unbonded areas

3.5 3 Adhesion of the shell edge binding after aging. When tested in accordance with 4.4 7, the edging shall not peel back more than 1/4 inch from the helmet shell

3 5 4 Adhesion of flange to earcup The bond of the flange to the earcup shall be such that when tested in accordance with 4 4 2 5 the flange does not separate from the earcup ir the area of the bond

3 5 5 Water immersion The external coating of the helmet shell shall show no evidence of softening, peeling, or blistering when tested as specified in 4.4.2.5.

3.5 6 Communication equipment operation. The communications equipment (headset-microphone assembly), after installation in the helmet, shall function satisfactorily when tested as specified in 4 4.7.

3 5 7 Sound attenuation. The minimum sound attenuating capability of the helmet shall be in accordance with the following values when tested as specified in 4 4 7.

Hertz units	125	250	500	1000	2000	3150	4000	6300	8000
Decibels	17	14	20	21	26	38	37	46	42

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3 5.8 Chinstrap/earcup retainer retention strength. When the chinstrap/ earcup retainer is tested as specified in 4.4.7, there shall be no separation of the snap fastener components, i.e., the separation of the male portion from the female portion, separation of the socket from the button or the eyelet from stud; there shall be no separation of the swivel fastener components; and the grommets and snap fasteners shall not pull out from the material.

3.5.9 Strength of components when procured separate from the helmet assembly.

3.5 9 1 Chinstrap. When tested as specified in 4.4.8, there shall be no separation of either of the snap fastener components, i.e., the button from the socket. The grommet and snap fasteners shall not pull out from the webbing, and the chinstrap shall not slip through the D-rings.

3.5.9.2 Earcup retainer. When tested as specified in 4.4 8, the snap fasteners provided for attaching the free end of the chinstrap, as identified on Drawing 8-2-833 shall not pull out from the fabric nor shall the snap fastener components separate, i.e., the stud from the eyelet Also the grommet for attaching the secured end of the chinstrap shall not pull out from the fabric.

3 5.9.3 Earcup retainer and chinstrap When earcup retainers and chinstraps are procured in the same contract, testing may be performed simultaneously by joining the two items as required on Drawings 8-2-800. The performance requirements for each item shall be as specified in 3.5.9.1 and 3.5.9.2.

3.5.10 Retention strength of eyelets in earcup retainer eyelet chapes When tested as specified in 4 4.2.5, the lot average of the retention strength of the eyelets in the earcup retainer eyelet chapes shall be not less than 100 pounds with no single determination less than 75 pounds.

3.6 Repairs. Except as otherwise specified herein, repairs are not allowed to be made to the helmet.

3 6.1 Shell repairs. Slight surface imperfections not greater than 1 ply deep, 1/8 inch wide, or 1/2 inch in length shall be filled. Shells having defects greater than 1 ply deep, 1/8 inch wide, or 1/2 inch long shall be scrapped.

3 7 Replacement of defective components. During the spreading, cutting and manufacturing process, textile, leather and polyurethane components having material defects, dimensional defects, or damages that are classified as defects in 4.4.2.3, 4.4.2.4, and 4.4.3 shall be removed from production and replaced with non-defective components.

3.8 Weight. The weight of a complete helmet assembly, without earcup spacer kit, extra visor and instruction manual shall not exceed 56 ounces.

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3.9 Dimensions of helmet shell. The dimensions of the helmet shell after edge trimming, after coating has dried, and prior to application of rubber edging shall be as specified in table I.

*TABLE I. Helmet shell dimensions (inches)

Characteristic	Size Small/Regular	X-Large
Length 1/	10 27 ± 0.09	10 61 ± 0 09
Overall width 2/	10 80 ± 0.09	11 45 ± 0.09
Edge width 3/	7.64 ± 0.13	8.12 ± 0 13
Diagonal difference 4/	Not greater than 0.06	Not greater than 0.06

1/ Length dimension is an outside dimension and shall be measured from the front center benchmark to the rear center benchmark. Measurement does not include the thickness of the benchmarks.

2/ Overall width is an outside dimension in the ear section and shall be taken between the points of maximum distance.

3/ Edge width is the outside distance between the bottom edge of the ear sections and shall be measured at the ear section benchmarks. Measurement does not include the thickness of the benchmarks.

4/ Diagonal difference is the difference between diagonal chord lengths measured at the outside edge from points 2 inches on either side of the front benchmarks diagonally to points 2 inches on either side of the rear benchmark.

3.10 Visor. The visor shall conform to MIL-V-43511 and Drawing 8-2-510. All testing shall be performed after the abrasion resistant coating has been applied

3.11 Workmanship. The finished helmet assembly shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4 1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.5 and tested for the characteristics specified in 4.4.7. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition, the components listed in table II shall be examined for all dimensions The sample unit shall be as designated in table II. The sample size for the designated lot size shall be as shown below The lot shall be unacceptable if one or more dimensions fail to meet the requirement on one or more sample units

Lot size	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

TABLE II. Inspection of components for dimensions

Component	Dimensional requirement	Number determinations per unit	Results reported numerically to nearest	Sample unit
Buckle	Dwg 8-2-855/856	1/	2/	1
Adapter	Dwg 8-2-821	1/	2/	1
Clip	Dwg 8-2-835	1/	2/	1
Hook	Dwg 8-2-834	1/	2/	1
Post 3/	Dwg 8-2-817/818	1/	2/	1
Visor lock assy 4/	Dwg 8-2-523	1/	2/	1
Spacer, visor housing	Dwg 8-2-481	1/	0.001, 1/64, half degree	1
Binding, shell edge	Dwg 8-2-800	1/	2/	One yard

1/ One determination shall be made for each specified dimension.

2/ Unless otherwise specified, results shall be reported to the nearest 0 001 inch, 1/64 inch, or one degree as applicable.

3/ Each size post shall be considered as a separate component.

4/ Each component of the visor lock assembly shall be considered as an individual component and shall be inspected individually.

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4.4.1.1 Certification. The contractor shall furnish a certificate of compliance certifying that all adhesives, materials, and components for which the requirements are not covered by a Federal Specification, Military Specification or Standard conform to the specified requirement. In addition, the contractor shall certify that the instruction manual cover and text paper conform to the requirements specified in 3.3 2.

* 4.4 2 In-process inspection Inspection of the helmet components during the manufacturing process to determine whether cut lengths and sizes of cut parts of the components listed below conform to the specified dimensional requirements. In addition, inspection shall be made during the manufacturing process to insure that holes prepunched in components for installation of eyelets, grommets, and snap fasteners are prepunched smaller than the outside diameter of the eyelet, grommet, or snap fastener barrel; that the holes are not punched in the setting operation, that benchmarks on helmet shells are present; and that adhesives where required are applied in a uniform manner Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the in-process or end item inspections are in accordance with the specified requirements. Whenever nonconformance is noted, corrections shall be made to the components affected and lot in process. Parts which cannot be corrected shall be removed from production.

Check for	Component	Drawing
Specified dimensions	Inner cushion, earcups	8-2-860
	Spacer, earcups	8-2-859
	Retainer, earcup	8-2-858
	Backing cushion, earcup	8-2-857
	Connector, earpad 1/	8-2-864
	Cover, earpad 1/	8-2-866
	Holding ring, earpad	8-2-864
	Cushion, earpad	8-2-862
	Button, earpad	8-2-863
	Pad, earcup chafing	8-2-827
	Earcup retainer reinforcement ring	8-2-840
Cut size or length	Body, earcup retainer	8-2-839
	Reinforcement, earcup retainer	8-2-839
	Extension, earcup retainer	8-2-841
	Binding, earcup retainer	8-2-812
	Nape strap, earcup retainer	8-2-812
	Chinstrap, helmet	8-2-815
	Strap, earcup tension cross strap	8-2-814

1/ Inspect only for dimensions which cannot be checked on the earpad assembly.

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4.4 2.1 Dimensional examination of initial production lot of molded or die formed helmet components except helmet shell Unless otherwise specified (see 6.2), the initial production lot of the items listed below shall be examined for all dimensional requirements specified on the applicable drawings. Any dimension exceeding the applicable tolerance shall be considered as a defect. The initial production lot of each component shall consist of an equal quantity of items made from each mold or die intended to be used in production of the items listed. Components required in different sizes and left and right hand shall be assembled in individual lots Unless otherwise specified in the contract or order, the minimum initial lot size for each component shall be 400 items (see 6.2). The sample unit for each lot of individual components shall be one for each component The samples selected for examination shall be randomly selected from each quantity of items made from each mold or die. The inspection levels and acceptable quality levels (AQLs), expressed in terms of defects per hundred units, shall be as indicated below

Component	Inspection level	AQL
Earcup	S-2	2.5
Flange, earcup	S-2	2.5
Earpad 1/	S-2	2.5
Spacer, tapered, visor housing	S-2	2.5
Tracks, visor	S-2	2.5
Housing, visor	S-1	6.5
Liner, helmet shell	S-1	6.5

1/ Applicable to only those dimensions which can be examined on the completed earpad assembly.

4 4.2.1 1 Dimensional examination of subsequent production lots of molded or die formed helmet components except helmet shell Except when changes are made to the contractor's manufacturing process or the material, mold, or die is changed, subsequent production lots of the components listed in 4 4.2.1 shall be dimensionally examined only for length, width, height, thickness, hole diameter, hole location, slot length and width as applicable. 1/ Any dimension exceeding the applicable tolerance shall be considered a defect If the contractor changes the manufacturing process or the material, mold, or die for any specific component, the initial production lot of the component involving the change shall be examined as specified in 4 4.2.1. Components required in different sizes and left and right hand shall be assembled in individual lots. The sample unit for each lot of individual components shall be one of each for each component. For each component the inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0

1/ For each subsequent production lot of each component, the contractor shall furnish a certificate of compliance certifying that the manufacturing process, materials, molds, or dies have not been changed from that used in the initial production lot.

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* 4.4.2.2 Mold, trim and hole authorization (see 6.2). Prior to beginning production under any contract, one helmet shell produced from each mold after edge trimming and curing sufficiently to allow the shell to stabilize (prior to painting and prior to application of rubber edging) shall be submitted to the procuring activity for examination for conformance to the applicable helmet shell Drawings 8-2-803, 8-2-804. Approval of the shell will authorize that mold for use in production under that contract. Any change to the mold, trimming process, drilling process, molding process, or materials shall require reauthorization.

4.4.2.2.1 Dimensional examination of helmet shell production lots. The helmet shell shall be examined to determine if the shell thickness 1/, hole locations, hole shapes, hole sizes, and overall dimensions conform to the dimensional requirements specified on the applicable drawings and in 3 9 The lot shall be expressed in units of size small, size medium, or size large helmet shells The sample unit shall be one helmet shell after edge trimming and curing sufficiently to allow the shell to stabilize (after coating has dried and prior to application of rubber edging). The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 2/

1/ The thickness of the helmet shell shall be measured at seven different locations. The area of the shell covered by the liner shall be divided into four quadrants. One measurement shall be taken at the approximate top center of the helmet and one measurement shall be taken at the approximate center of each of the four quadrants. One measurement shall be taken in each one of the ear sections within 1/2 inch of the lower edge adjacent to the benchmarks. Any measurement found to be less than 0.060 inches (0.048 inches in the ear section) or greater than .095 inches (0.080 inches in the ear section) shall be classified as a defect.

2/ For each production lot of helmet shells, the contractor shall furnish a certificate of compliance certifying that the manufacturing process, materials, and molds have not been changed from those authorized in 4.4.2.2.

* 4.4.2.3 Visual examination of form fit suspension assembly, earcup retainer, and chinstrap assembly. The The form fit suspension assembly (Drawing 8-2-897), earcup retainer assembly (Drawing 8-2-812), and chinstrap assembly (Drawing 8-2-815) shall be examined for visual defects prior to installation into the helmet shell. The lot size shall be expressed in units of form fit suspension assemblies, earcup retainer assemblies, and chin strap assemblies, as applicable. The sample unit for each lot of the individual items shall be one for each item. For each item, the inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total (major and minor combined) defects. Any critical defect found during sampling inspection for major and total defects shall be cause for rejection of the lot represented by the sample. The defects shall be classified in accordance with the list below.

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Examine	Defect	Classification Critical	Major	Minor
Webbing (nylon and elastic)	Any hole, cut, or tear· (applicable nylon webbing)			
	- Up to 1/4 inch in length		X	
	- More than 1/4 inch in length	X		
	Any hole, cut, or tear in elastic webbing		X	
	Edges scalloped, not firmly and tightly woven, or frayed edge			X
	Ends not fused or dipped as required			X
	Needle chews:			
	- Up to 1/8 inch in length			X
	- More than 1/8 inch in length		X	
Cloth	Any hole, cut, or tear:			
	- Up to 1/4 inch in length		X	
	- More than 1/4 inch in length	X		
	Any abrasion marks, broken or missing yarns, or multiple floats			X
	Any mend, darn, or patch		X	
	Any raw edge		X	
	Any needle chew		X	
Polyurethane foam	Any cut, hole, deep wrinkle, tear, or splice		X	
Hardware, general	Broken or malformed	X		
	Corroded areas, burrs, or sharp edges		X	
	Finish omitted or not as specified, where required		X	
	Any required component improperly installed causing failure to serve intended purpose	X		
Snap fastener	Clinched excessively tight, cutting adjacent material		X	
	Clinched loosely to the degree that components can be expected to become detached during use, i.e., components rotate freely, ends of barrels of button and eyelet components not completely rolled back	X		
	Incorrect style	X		

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Examine	Defect	Classification Critical	Major	Minor
Eyelets and grommets	Clinched excessively tight, cutting material or causing distortion to the degree that mounting hardware will not pass through hole		X	
	Insecurely clinched to the degree that eyelet or grommet may be detached from material	X		
	Grommet insecurely clinched to the degree that grommet can be rotated freely within the material	X		
	Any split in the eyelet or grommet		X	
Open seams	One-half inch or less			X
	More than 1/2 inch: - On seams around periphery of openings for earcups (two seams each opening) and seams for securing grommet and snap fastener reinforcements (applicable earcup retainer)	X		
	- On all seams other than above		X	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur.			
Runoffs (see open seams)				
Seams and stitch type	Wrong seam or stitch type		X	
Stitch tension	Loose, resulting in loose top or bobbin thread, i.e., thread does not lie flush with material			X
	Tight resulting in puckering of material			X

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Examine	Defect	Classification Critical	Major	Minor
Stitches per inch	One less than minimum specified			X
	Two or more stitches less than minimum specified		X	
	One or more stitches in excess of maximum specified			X
Thread breaks, skipped stitches or runoffs	Overstitched less than 1/2 inch in each direction beyond defective stitching area			X
NOTE:	Thread breaks or two or more consecutive skipped or runoff stitches not overstitched shall be classified as open seams			
Stitching ends	Not secured as specified			X
Rows of stitching	Any row missing except on box-x stitching		X	
Box-x stitching	Around periphery of stitch pattern:			
	- Row omitted where single row is required or both rows omitted where double row is required		X	
	- Row omitted where double row is required			X
	Diagonal row or rows omitted			X
Bartacks	Any omitted		X	
	Stitching loose, incomplete, or broken.			
	- On earcup retainer eyelet chapes	X		
	- On bartacks other than above		X	
Components and assembly	Any required component or operation omitted (unless otherwise classified herein)	X		
	Any component misplaced or not assembled as specified		X	
	Number of folds at ends of webbing, where required, less than number specified			X

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Examine	Defect	Classification Critical	Major	Minor
Binding	Loosely applied but not exposing raw edge of material			X
	Loosely applied exposing raw raw edge of material		X	
Cleanness	Grease or oil stains clearly noticeable; thread ends not trimmed throughout as specified			X
Color of textile components	Color of any textile component not as specified			X
Identification markings (applicable when components are procured separately)	Omitted, incorrect, illegible, or location not as specified			X

4.4.2.4 Dimensional examination of earcup retainer assembly. The earcup retainer assembly (Drawing 8-2-812) shall be examined for dimensional defects prior to installation into the helmet shell. The lot size shall be expressed in units of earcup retainer assemblies. The sample unit shall be one earcup retainer assembly. The earcup retainer shall be examined for all dimensions shown on the applicable drawing including the hole diameter of the eyelets and position of eyelet in eyelet tab. Any dimension exceeding the applicable tolerance shall constitute a defect. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 10.0.

4 4.2.5 In-process testing of helmet components The helmet components shall be tested for the characteristics listed in table III. The sample unit for each lot of individual components shall be one for each component. For all tests, the inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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TABLE III. Helmet components in-process tests

Characteristic	Requirement para.	Test method	No. of deter. per sample unit	Results reported as
Adhesion of flange to earcup	3 5 4	4.5 2	1	Pass or fail
Water immersion 1/	3 5 5	4.5 3	1	Pass or fail
Retention strength of eyelets in earcup retainer eyelet chapes 2/	3 5 10	4.5.13	1	Pass or fail

1/ Helmet shells which pass the water immersion test may be returned to production.

2/ Testing of the eyelet chapes is required prior to attaching the chapes to the earcup retainer body

4 4.3 End item visual examination. The helmet assembly complete with extra visor, earcup spacer kit, and instruction manual shall be examined for the defects listed below The lot size shall be expressed in units of complete helmet assemblies The sample unit shall be one complete helmet assembly. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10 0 for total (major and minor combined) defects Any critical defect found during sampling inspection for major and total defects shall be cause for rejection of the lot represented by the sample.

Examine	Defect	Classification Critical	Major	Minor
Exterior surface of shell	Any delamination, tear, cut, hole, crack, dent, nick, or gouge	X		
	Not smooth (except at benchmarks)			X
	Any pinhole			X
	Any pimple			X
	Any unauthorized repair	X		

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*Examine	Defect	Classification Critical	Major	Minor
Exterior finish coating of shell	Omitted		X	
	Thin film (shell showing through)			X
	Finish not applied prior to attachment of other hardware		X	
	Coating comes off underneath screw heads (two screws shall be removed to accomplish this inspection)			X
	Finish marred or partially removed by adhesives applied to helmet components			X
	Any orange peel, run, or sag			X
	Tacky surface			X
	Any ridge or streak (except at benchmarks)			X
	Any scuffed area or scratch			X
	Color not uniform or not olive drab			X
	Poor adhesion (furrows or flakes off when scratched with fingernail)			X
	Any dirt, grease, stain, or paint smudge			X
Visor tracks and tapered spacers	Any rough edge			X
	Any crack			X
	More than one spacer used under each visor track			X
Visor lock assembly	Plastic knob not olive drab			X
	Plastic knob cracked		X	
	Does not function properly or does not slide freely in visor housing slot		X	
	NOTE: To check function of visor lock, loosen, position, and tighten the lock in the full up and down position. The lock shall securely hold the visor in the set position and the lock shall freely slide in the visor housing slot without binding.			

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Examine	Defect	Classification Critical	Major	Minor
Visor lock assembly (cont'd)	Lock assembly cannot be disengaged from visor and visor housing		X	
Visor housing	Cracked or chipped or insufficient clearance between housing and visor	X		
	Distorted, causing visor to touch housing at any point 3/4 inch or more beyond left or right edge of visor (To be examined while moving visor to full down and up position)	X		
	Front edge projects below edge of shell:			
	- 1/16 inch up to 1/8 inch			X
	- more than 1/8 inch		X	
Visors (applicable to installed and spare visor)	Crack, hole, chip, or break	X		
	Evidence of warpage	X		
	Surface abraded, scratched, or dulled	X		
	Stain, paint, or discoloration not readily removed with water	X		
	Surface not smooth	X		
	Edge lacks required radius		X	
	Any bonding gap in visor/lens guide common surface		X	
	Any bonding buildup, on lens guide or visor			X
	NOTE. Adhesives are unauthorized.			
	Visor does not freely slide in visor tracks (to be examined while moving visor to full down and up position)		X	
	Installed visor not neutral grey			X
	Spare visor not clear			X

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*Examine	Defect	Classification Critical	Major	Minor
Shell edge binding	Any cut, tear, or hole		X	
	Not fully positioned over edge of helmet shell in any specific area and 1/4 inch or longer in length:			
	- 1/16 inch up to 1/8 inch			X
	- more than 1/8 inch		X	
	Any area not adhered to shell greater than 1/2 inch in length and 1/8 inch in width		X	
	Several unadhered areas the total length of which are 2 inches or more in length or any two unadhered areas less than 1/2 inch apart			X
	Adhesive extends beyond edge of binding more than 1/16 inch for more than 1/4 inch at one or more places			X
	Ends not butted together:			
	- Open 1/16 inch up to 1/8 inch			X
	- Open more than 1/8 inch		X	
	Butt joint not in rear of helmet			X
	Not one piece			X
Shell liner	Not one piece	X		
	Not fastened		X	
	Front edge of liner not abutted against edge binding or not located centrally as specified in Drawings 8-2-800			X
	Any patched area (voids or tears filled, except over retainer screws)			X
	Any hole, tear, crack, or partially compressed area	X		
	Surface not smooth			X
	Missing hook fasteners	X		
Form fit suspension assembly	Any hole, cut, tear or frayed edge on the cloth cover		X	
	Cloth cover improperly installed on plastic layers			X
	Assembly severely misshapen or malformed	X		

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Examine	Defect	Classification Critical	Major	Minor
Earcup assembly: (Examine components with earpad, earphone and earcup cushion removed)				
Earcup and earcup flange	Cracked, chipped, broken, or dented		X	
	Surface not smooth			X
	Flange not securely adhered to earcup around entire periphery (check by applying force to flange with hand)		X	
	Backing cushion:			
	- Omitted		X	
	- Cut, tear, or hole		X	
	- Not adhered to earcup as specified		X	
Earcup cushion	Any component omitted		X	
	Any component not completely adhered leaving open area greater than 1/2 inch in length and 1/8 inch in depth		X	
	Any cut, tear, hole, or deep wrinkle		X	
Earpad	Any heat seal not complete		X	
	Distorted due to misalignment of components during heat sealing operations		X	
	Any hole (except vent hole)		X	
	Any cut, blister, bubble, tear, abraded area, or crack		X	
	Not smooth			X
	Vent holes mislocated or not in the quantity specified			X

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Examine	Defect	Classification Critical	Major	Minor
Chinstrap and earcup tension cross straps	Any hole, cut, tear, or frayed edge		X	
	Scalloped edges, not firmly and tightly woven			X
	Ends of cross straps not dipped as required			X
	Needle chews:			
	- Up to 1/8 inch in length			X
	- More than 1/8 inch in length		X	
	Hardware broken or malformed:			
	- On chinstrap	X		
	- On cross straps		X	
	Corroded areas, burrs, or sharp edges		X	
	Finish on hardware, where required, omitted or not as specified			X
	Rivet on chinstrap swivel fastener not peened as specified		X	
	Any required hardware component improperly installed causing failure to serve intended purpose:			
	- On chinstrap	X		
	- On cross straps		X	
	On chinstrap, end of adjustment strap at swivel fastener end not folded to the outside as shown on Drawing 8-2-815		X	
Snap fastener	Any fastener not functioning properly, i.e., fails to snap closed, provide a secure closure or to open freely	X		
	NOTE: The chinstrap shall be snapped and unsnapped twice to the earcup retainer assembly to determine whether the parts of the fastener fully close, provide a secure closure and separate freely.			

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Examine	Defect	Classification Critical	Major	Minor
Snap fastener (cont'd)	Clinched excessively tight, cutting material		X	
	Clinched loosely to the degree that components can be expected to become detached during use, i e , components rotate freely, ends of button and eyelet components not completely rolled back	X		
	Incorrect style	X		
	Orientation of button and socket not as specified	X		
Grommets (applicable to chinstrap)	Clinched excessively tight cutting webbing		X	
	Insecurely clinched to the degree that grommet can be rotated freely within the webbing or can be detached from the webbing	X		
Seams and stitching (applicable to chinstrap and earcup tension cross straps)	Stitching defects shall be scored same as specified in 4.4.2.3			
Earcup chafing pad	Any cut, hole, tear, or deep wrinkle		X	
	Not completely adhered to helmet shell leaving open area greater than 1/2 inch in length and 1/8 inch in depth		X	
	Not covering microphone boom attachment hardware or the lower cross strap posts		X	
Earcup retainer assembly, general	Any component of the earcup retainer not attached to helmet shell in specified location or orientatation		X	
	Earcups not securely retained by earcup retainer		X	
	Earcup tension cross straps:			
	- Any omitted		X	
	- Any not attached as specified			X

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Examine	Defect	Classification Critical	Major	Minor
Mounting hardware (screws, posts, and washers)	Any screw not securely tightened		X	
	Any screw too tight, causing cutting or crushing of shell		X	
	Washers omitted			X
	Thread sealant omitted on any screw		X	
	Any burr or sharp edge		X	
	Not finished (when applicable)			X
	Corroded			X
Retainer	Screws for attaching earcup assembly to helmet:			
	- Any screw end protrudes more than 1/32 inch but not more than 3/64 inch from surface of clip or post (as applicable)			X
	- Any screw end protrudes more than 3/64 inch from surface of clip or post (as applicable)		X	
	- Any screw end less than flush with surface of clip by 1/64 inch or more (not applicable) to post		X	
Headset-microphone assembly.				
General	Any wire cut, cracked, or abraded		X	
	Any hardware component:			
	- Corroded			X
	- Omitted		X	
	- Not attached as specified		X	
	Any component omitted		X	
	Grommet in shell inclosing microphone lead not fully seated			X
Earphones	Wires not routed as specified		X	
	Grommets inclosing earphone wires in earcup not fully seated		X	
	Set screws not securely tightened in earphones		X	
	Dirt or any foreign matter in earphone baffle holes		X	

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Examine	Defect	Classification Critical	Major	Minor
Microphone	Cracked		X	
	Not adjustable by hand		X	
Microphone boom	Broken or deformed		X	
	Will not adjust or swing when locking screw is loosened, or cannot be locked in set position		X	
	Boom washer not between shell and earcup chafing pad		X	
Components and assembly - general	Any operation not performed properly		X	
	Any component omitted or not attached as specified unless otherwise classified herein		X	
	Color of any component not as specified			X
	Any hardware component not finished as specified unless otherwise classified herein			X
Spacer kit	Not packaged as specified			X
	Any component omitted		X	
Identification marking	Omitted, incorrect, or illegible			X
	Not firmly attached or location not as specified			X
Instruction manual	Omitted, incorrect, or figures illegible, or any page omitted			X
Cleanness	Any grease, oil, or stain clearly noticeable			X

4.4.4 End item dimensional examination. The chinstrap and earcup tension cross strap shall be examined for conformance to the dimensions specified on Drawings 8-2-815 and 8-2-814. The location of the liner within the helmet shell shall be examined for conformance with the requirements specified on Drawing 8-2-800 Any nonconformance shall be classified as a defect. The lot size shall be expressed in units of helmet assemblies. The sample unit shall be one complete helmet assembly. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4 4.5 Weight examination of the completed helmet assembly. The complete helmet assembly without earcup spacer kit, instruction booklet, and extra visor shall be examined for conformance to the weight requirement of 3.8. The lot size shall be expressed in units of helmet assemblies The sample unit shall be one complete helmet assembly. Defects shall be classified in accordance with the list below. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 4.0 for total defects.

Examine	Defect	Classification Major	Minor
Weight 1/	More than specified·		
	- Up to 1 ounce		X
	- By more than 1 ounce	X	

1/ Weighing of the helmet shall be conducted on a balance or scale graduated in increments of 1/4 ounce minimum (a scale or balance graduated in grams may be used and the results converted to ounces).

4.4.6 End item critical defects examination. Every end item found acceptable under the visual and dimensional examinations in 4.4.3 and 4 4.4 shall be examined for the critical defects listed in 4.4.3. Any end item found to contain one or more critical defects shall be rejected.

4.4.7 End item testing. The complete helmet assembly shall be tested for the characteristics listed in table IV The inspection lot shall consist of helmets of one size only. The lot size shall be expressed in units of one helmet. The sample unit shall be as indicated in table IV. The helmets used for impact testing shall afterwards be used for the chinstrap/earcup retainer retention strength test. The helmets used for impact testing at ambient conditions may also be used afterwards for the edge binding tests Tests for impact resistance and sound attenuation shall be performed only in laboratories certified by the Government as acceptable for the performance of those tests (see 6.5).

4.4.7.1 Impact resistance testing. The sample size for the designated lot size shall be as indicated below. Failure of one or more individual impact tests (twenty total required per sample unit) shall be cause for rejection of the lot.

Lot size	Sample size (sample units)
400 or less	1
401 up to 800	2
801 up to 1200	3
1201 up to 1600	4
1601 up to 2000	5

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For lot sizes greater than 2000 the sample size shall be increased by one for each additional quantity of 400 items, e.g., for a lot of 2400 items the sample size shall be six.

4 4 7 2 Shell edge binding adhesion testing. For edge binding testing the inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4 7.3 Communication equipment operation testing One hundred percent of the helmet assemblies in the lot shall be tested for operation of the communication equipment. Any helmet failing to pass the test shall be rejected.

4.4 7 4 Sound attenuation testing. One percent of the lot shall be tested for sound attenuation Failure of one or more helmets to pass the test shall be cause for rejection of the lot

4 4 7.5 Chinstrap/earcup retainer retention strength testing. Each helmet to be subjected to impact testing shall be tested for chinstrap/earcup retainer retention strength. The test shall be performed after conditioning the impacted helmets for a minimum of 8 hours at standard ambient temperature. Failure of one or more helmets to pass the test shall be cause for rejection of the lot

TABLE IV. Helmet assembly tests

Characteristic	Requirement para	Test method	No deter per unit	Results reported as pass or fail	Sample unit
Impact resistance 1/	3 5.1	4 5.4	20	X 2/	4 helmets
Shell edge binding adhesion before aging	3.5.2	4.5.5	1	X	1 helmet
Shell edge binding adhesion after aging	3.5.3	4.5.6	4	X	1 helmet
Communications equipment operation	3.5.6	4.5.7	1	X	1 helmet
Sound attenuation	3.5.7	4.5.8	1	X 2/	1 helmet
Chinstrap-earcup retainer retention strength	3.5.8	4 5.9	4	X	4 helmets

1/ A different helmet shall be used for impact testing for each of the four exposure conditions.

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2/ All test reports shall contain the individual values utilized in expressing the final results.

4.4.8 Testing of components procured separate from the helmet assembly. The components listed in table V, when procured separate from the helmet assembly, shall be tested for the applicable requirement. The lot shall be expressed in units of size small, size medium, and size large earcup retainers or chinstraps. The sample unit shall be as indicated in table V. The inspection level shall be S-1. Any test failure shall be cause for rejection of the lot .

TABLE V. Instruction for testing of components when procured separately from the helmet assembly

Component	Requirement para.	Test method	Results reported as pass or fail	Sample unit
Chinstraps 1/	3.5.9.1	4.5.10	X	1
Earcup retainer 1/	3 5.9.2	4.5.11	X	1
Earcup retainer and chinstrap 1/	3 5.9.3	4.5.12	X 2/	3/

1/ All tested items are unserviceable and shall not be retained in the lot.

2/ Testing of the earcup retainer and chinstrap simultaneously is optional, (see 3.5.9.3). If performed in this manner and one component fails one or more of the strength tests, the other component shall be considered as having been subjected to an incomplete test and shall be retested. Four determinations shall be made: (1) Strength of snap fasteners and swivel fastener on chinstraps, (2) strength of snap fasteners on earcup retainer, (3) strength of grommet on earcup retainer, and (4) strength of grommet on chinstrap.

3/ The sample unit shall be one earcup retainer and one chinstrap joined as specified in 3.5.9.3. For the purpose of ascertaining acceptance of the items the earcup retainer shall be considered as representative of a lot of earcup retainers and the chinstrap shall be considered as representative of a lot of chinstraps.

4.4.9 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing. Any component damaged or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of helmets or accessories per container is more or less than required

4.4 10 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified Load not bonded with required straps or film as specified.
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 Test conditions. Unless otherwise specified, all tests shall be performed at 73° ± 5°F and 50 percent ± 10 percent relative humidity and test samples shall be exposed to these conditions for at least 24 hours prior to test.

4.5.2 Adhesion of flange to earcup test. With the earcup restrained in a vise or other suitable apparatus, apply sufficient force to the flange along its entire outer peripheral edge to cause separation of the flange from the earcup. Separation of the flange from the earcup at the bond line shall constitute failure of the test. Breaking of the flange or earcup is the acceptable condition.

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4 5.3 Water immersion test. The helmet shell shall be immersed in tap water at 60° to 80°F for 16 hours minimum. At the expiration of this time, the helmet shell shall be removed and allowed to dry by air drying or by forced air drying under the same temperature conditions. The exterior coating of the helmet shell shall be examined and any evidence of softening, blistering, or peeling of the finish shall constitute failure of the test.

* 4.5.4 Impact test. The helmet shall be impact tested after exposure to ambient temperature, low temperature, high temperature, and water immersion in accordance with paragraphs 7.2, 8 2 through 8.6, and 9 and all other applicable paragraphs of American National Standard Z90.1b-1979 with the following deviations and modifications:

a. Paragraph 9 3.1, delete contents and substitute the following:

Each helmet tested shall be impacted with a single impact at each of five impact sites: rear, front, left side, right side, and top. The impact sites are to be determined as follows: Rear and front· orient the flat base of the headform to 20 plus or minus 1.0 degree from the vertical; left and right sides: orient the flat base of the headform to 35 plus or minus 2 0 degrees from the vertical; top (apex): orient the flat base of the headform to 10 plus or minus 1.0 degree from the horizontal, pitched rearward. The first site to be impacted shall be the rear of the helmet. No impact site is to be nearer than 3.5 inches from any other impact site.

b. Paragraph 9.3.2, delete contents and substitute the following:

The anvil used for all impact tests shall be a flat steel plate no less than 1.0 inch thick and 5.0 inches in diameter.

c. Paragraph 9.3.3, delete contents and substitute the following:

The helmet shall be dropped from a height sufficient to produce an impact velocity (terminal velocity) of 19.6 plus 0.3, minus 0.0, feet per second.

d. Paragraph 12.2, add the following:

(1) The instrumentation system shall be calibrated at least once a year by and independent agency. The calibration must be dynamic and traceable to the National Bureau of Standards.

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(2) If a test series is interrupted for more than 3 hours, prior to re-starting the test series at least three calibration impacts shall be performed.

The test shall be considered a failure if one or more individual impact tests exceeds the fail criteria specified in paragraph 9.2 of Z90.1b-1979.

4.5.5 Adhesion of edge binding to helmet shell (before aging) test. The adhesion of the binding to the helmet shell before aging shall be tested by manual and visual inspection of the edges of the binding on both the outside and inside of the helmet shell around its entire periphery The binding shall be considered unbonded when the edge can be rolled back on itself and away from the shell by applying force with the ball of the thumb. If the unbonded areas exceed the limitations specified in 3.5.2, the test shall be considered as a failure

4 5.6 Adhesion of shell edge binding to helmet shell (after aging) test. The helmet shall be placed on its crown in a circulating air oven at 160° ± 5°F for 4 hours minimum but not more than 24 hours. The helmet shall then be removed from the oven and conditioned at the standard conditions specified in 4.5.1. After conditioning, the adhesion of the edge binding shall be tested as follows:

a The adhesion test shall be performed in two separate areas, one at the bottom of one of the ear sections and the other in the rear of the helmet.

b Slice off the rounded portion of the edge binding with a sharp knife to expose the edge of the shell for a distance of 2 inches minimum. At one end of the cut, sever the remaining sides of the edge binding. Peel back the edge binding along the cut on both the inside and outside to form tabs 1/2 + 1/8 - 0 inch in length.

c To one of the tabs, attach a 0.25 pound weight. Position and secure the helmet so that the weight pulls at a right angle to the plane of the bond line to be tested Make suitable gage marks on the edge binding and shell for the measurement of the amount of peel back after completion of the test. Suspend the weight for 1 hour minimum and measure the amount of peel back of the tab to the nearest 1/16 inch. Test the other tab in the same manner

d The test shall be considered a failure if one or more of the tabs peels back in excess of the limitation specified in 3 5.3.

4.5.7 Communications equipment (talk-listen test). Each helmet assembly shall be tested by connecting the headset-microphone assembly to a suitable test system. The presence of sidetone in the headset, when the microphone unit is spoken into, constitutes the test. Failure to hear sidetone in either earphone of the headset shall be evidence of failure of the communications equipment to properly perform.

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4.5.8 Sound attenuation test. Sound attenuation of the helmet shall be determined in accordance with American National Standards Institute Method S12.6/ASA 55 Failure of the helmet to meet any of the values specified in 3.5.7 shall constitute failure of the test

4.5.9 Chinstrap/earcup retainer retention strength test. The helmet shall be mounted for testing chinstrap/earcup retainer retention strength in accor-dance with paragraph 11.1 of American National Standard No. Z90.1b-1979. A 300 pound weight or tension equivalent thereto shall be gradually applied to the device retained by the chinstrap for not less than 2 minutes. Failure of the chinstrap/earcup retainer to meet one or more of the performance requirements specified in 3.5.8 shall constitute failure of the test. If a failure occurs and it is ascertained that the failure may have occurred as a result of damage to the chinstrap/earcup retainer assembly due to stresses applied during impact testing of the helmet, then the test shall be repeated using another helmet.

4.5.10 Strength of chinstrap (when procured separate from helmet assembly) test. The chinstrap, when procured separate from the helmet assembly shall be pull tested at 200 pounds minimum to test the strength of the snap fasteners, grommet and swivel fastener The load shall be applied at the rate of 20 inches per minute. The test set-up shall be as shown on Drawing 8-2-825 Failure of the chinstrap to meet one or more of the performance requirements specified in 3.5.9.1 shall constitute failure of the test.

4.5.11 Strength of earcup retainer (when procured separately from the helmet assembly) test The earcup retainer, when procured separate from the helmet assembly,shall be pulled tested to 200 pounds minimum to test the strength of the snap fasteners provided for attaching the free end of the chinstrap and also the grommet for attaching the secured end of the chinstrap. The load shall be applied at the rate of 20 inches per minute The test set-up for each test shall be as shown on Drawing 8-2-833. Failure of the earcup retainer to meet one or more of the performance requirements specified in 3.5 9 2 shall constitute failure of the test.

* 4 5.12 Strength of earcup retainer and chinstrap (when procured separate from the helmet assembly and tested simultaneously) test. The earcup retainer and chinstrap, when procured separate from the helmet assembly and tested simulta-neously, shall be tested without the use of test straps. All testing shall be performed with the two components joined as specified in Drawing 8-2-800. The clamps shall be attached to the components in the same manner as shown on Drawing 8-2-833 when using the test straps. The force and rate of application shall be the same as specified in 4.5.10 and 4.5.11. Failure of the earcup retainer to meet one or more of the performance requirements specified in 3.5.9.2 shall constitute failure of the earcup retainer portion of the test. Failure of the chinstrap to meet one or more of the performance requirements specified in 3.5.9.1 shall constitute failure of the chinstrap portion of the test.

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4.5.13 Retention strength test of eyelets in earcup retainer eyelet chapes. The earcup retainer eyelet chapes shall be pull tested for the retention strength of eyelets within the chape webbing. The test set-up shall be as shown on Drawing 8-2-832 The load shall be applied at the rate of 20 inches per minute The individual values and the lot average shall be reported Failure of the eyelet chapes to meet the requirements specified in 3.5 10 shall constitute failure of the test.

5. PACKAGING

5 1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2)

5 1 1 Level A. Each completely equipped helmet with accessories shall be preserved in accordance with Method IC-3 of MIL-P-116 and detailed as hereinafter specified. Cushioning material may be utilized to separate the visor housing and visor to prevent contact. The waterproof bag shall be fabricated from L-P-378 material. Excess air shall be removed from each bag before final closure The microphone boom shall be fully extended with the microphone in a horizontal position in the open area at the front of the helmet The microphone shall be completely inclosed in minimum 1/4-inch thickness of paperbacked cellulosic cushioning material and secured in place with pressure-sensitive tape. The microphone plug and cord, instruction manual, and the earcup spacer kit shall be placed within the helmet. The spare visor shall be inserted in a cotton flannel bag, overwrapped in at least three thicknesses of tissue paper, and then placed within the helmet under the plug and cord The chinstrap shall then be snapped in place Each helmet shall then be unit packed in a fiberboard box provided with a one-piece die-cut fiberboard insert as illustrated on Drawing 8-2-524. The fiberboard box shall conform to style CSSC, grade W6c, or V3c of PPP-B-636. The fiberboard insert shall be fabricated from the same material as the fiberboard box. Inside dimensions of the box shall approximate 12-3/4 inches in length, 12-3/4 inches in width, and 11 inches in depth. Approximate dimensions are furnished as a guide only The top and bottom flaps of each box shall be secured with 2-inch minimum width pressure-sensitive waterproof or water-resistant tape conforming to PPP-T-60 or PPP-T-76.

5.1 2 Commercial. The helmets shall be preserved in accordance with ASTM D 3951

5 2 Packing. Packing shall be level A, B, or Commercial as specified (see 6 2)

5.2.1 Level A packing. Six helmets of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style FOL, grade V2s of PPP-B-636. The unit packs shall be packed flat, right side up, three in length, two in width, and one in depth within a shipping container. Inside dimensions of the shipping container shall approximate 39-1/2 inches in length, 26-1/2 inches in width, and 11-1/2 inches in depth Approximate dimensions are

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furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of the container specification except that the inspection shall be in accordance with 4.4.9. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6 2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads

5.2.2 Level B packing. Six helmets of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style OSC (minimum 3-inch overlap), type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The unit packs shall be packed flat, three in length, two in width, and one in depth within a shipping container. Inside dimensions of the shipping container shall approximate 39-1/4 inches in length, 26-1/2 inches in width, and 11-1/2 inches in depth Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.4.9.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with method III as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.4.9.

5.2.3 Commercial packing. Helmets, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), helmets, packed as specified in 5.2.2 and 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O and P. Pallet pattern shall be number 89 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Special marking. Arrows or prominent marks shall be placed on two sides of each unit pack and the shipping container to indicate the position in which the container shall be handled in transit or storage. The word "UP" shall be paced at the point of the arrows. The letters and arrows shall be at least 2 inches in height. The side panels of each interior package and shipping container shall be plainly marked as follows:

UP HANDLE WITH CARE UP

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The top panel shall be plainly marked as follows:

HANDLE WITH CARE

TOP

THIS SIDE UP

6. NOTES

6.1 Intended use. The helmet is for use by aviation personnel.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Size of helmet required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.4).
d. When initial production lot examination is not required (see 4.4.2.1).
e. If size of initial production lot of items listed in 4.4.2.1 is other than specified.
f. Mold, trim and hole authorization (see 4.4.2.2).
g. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
h. Type and class of unit load required (see 5.2.1).
i. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
j. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents, regarding arrangements for selection, inspection, and approval of the first article.

6.5 Certification of sound attenuation and impact resistance testing laboratories. Certification may be obtained from the U.S. Army Aeromedical Research Laboratory, Fort Rucker, AL 36362. Certification will be based on comparison of test results from the contractor's laboratory with the results of verification tests performed by the Aeromedical Research Laboratory and by inspection of contractor's laboratory facilities and procedures if considered necessary.

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6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document or drawing (see 3.3).

6.7 Supersession data. The type II helmet has been deleted from this document as it is no longer required.

6.8 Subject term (key word) listing.

Flight helmets
Headgear
Helmet

6.9 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

Project No. 8415-0657

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-H-43925D	Helmet, Flyer's, Protective, SPH-4

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

NATICK OP-1, Feb 86

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